Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production engineering experience building Python automation, web applications, and AWS cloud platforms (Lambda, EC2, S3, RDS) with MySQL, across Agile SDLC teams. Current research in LLM-powered testing pipelines.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, C/C++, SQL, Go.

Web & Backend: Django, Flask, FastAPI, React, Node.js, HTML/CSS, REST APIs.

Databases: MySQL, PostgreSQL.

Developer Tools: AWS (Lambda, EC2, S3, RDS), Git, GitLab CI/CD, Jenkins, Docker, Kubernetes, Terraform, Linux, Bash, Agile/Scrum, Jira, Confluence, Slack.

AI & Data: Gemini, Vertex AI, PyTorch, scikit-learn, LangChain, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity, supporting functional-testing verification workflows.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3) on AWS, replacing legacy Ruby scripts and reducing deployment time by **40%** across cloud environments.
- Executed **44 production environment upgrades** in 4 months on Agile Release Train, diagnosing live infrastructure issues across Linux, EC2, and databases.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** web application that digitized departmental record management, automated approval workflows, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, FastAPI, PostgreSQL, scikit-learn] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset) with FastAPI REST API and PostgreSQL/Redis, deployed via Docker on Oracle Cloud; cut RMSE **43%** over baseline Random Forest, improving weighted MAPE from **18.77% to 16.52%**.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).